

Aim 4: Integrate the social, economic drivers of variation in fishing pressure into a model projecting fishery sustainability and ecosystem services under different climate scenarios.

We will adapt and extend an existing model of the ecological dynamics of summer flounder and angler which was developed by co-PI Fay for the Management Strategy Evaluation (MSE; Bunnefeld et al. 2011, Punt 2017) of the summer flounder fishery that was completed in 2022 by the MAFMC, as part of their Ecosystem Approach to Fisheries Management risk assessment and structured decision process (Gaichas et al. 2019, Muffley et al. 2021, DePiper et al. 2021). The MSE uses a co-production approach to combine stakeholder engagement activities to define goals and objectives, policy alternatives, and plausible system scenarios, with simulation models of the full management cycle. MSE models are designed to include ecosystem dynamics, information from scientific surveys and landings, assessment, and management recommendations, which are then fed back to the simulated system dynamics. Thus, the MSE approach can incorporate and test uncertainties throughout a socio-environmental system. **Adapting the MSE co-production model completed in 2022 to include the processes identified in Aims 1-3 will allow us to refine and test our specific hypotheses regarding the drivers of socio-environmental dynamics in the summer flounder fishery and produce generalizable insights reflecting real-world conditions into the cross-disciplinary mechanisms that promote or undermine sustainability of coastal fisheries.**

Preliminary work - The goal of the initial MSE was to compare options designed to reduce discarding in the recreational summer flounder fishery (MAFMC 2022). In a collaborative, stakeholder-driven, and science-based process, co-PI Fay developed a modeling framework that integrated a full summer flounder population dynamics model with an angler behavior model to understand how the recreational sector at the state level responds to changing regulations and the climate-driven shift in geographic distribution (MAFMC 2022, Fay 2022). A key goal was to understand how stringent regulations on anglers (larger minimum sizes) can inadvertently increase recreational fishing mortality (Fig. 10). In each simulation, an operating model, representing the population dynamics of the summer flounder stock, its response to fishing, and the dynamics of the recreational fishery, is projected forwards in time by applying a management model that emulates the results of scientific assessments, incorporating uncertainty.

Proposed research design and analysis of different scenarios to test specific hypotheses: Fay and Kindsvater will work with a postdoc and a PhD student to integrate results from Aims 1-3 into and explicitly address components of the summer flounder fishery system that were either not included in previous work or were incorporated using simplified assumptions. The full integrated model can then be parametrized for different scenarios of management, climate, and social dynamics.

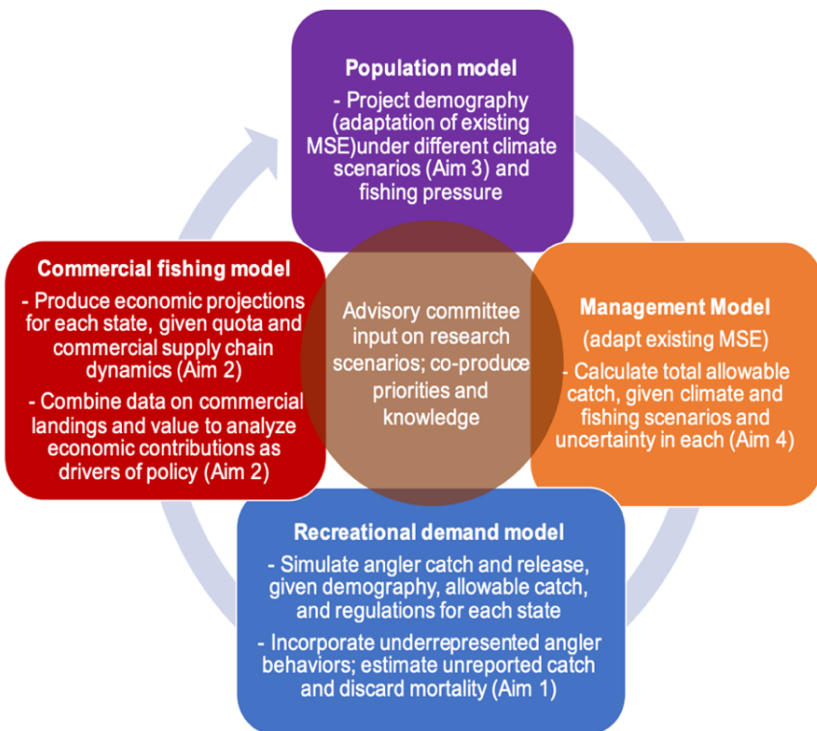


Fig 10: Overview of sequence of submodels that represent processes that will be incorporated in the proposed MSE.

Running the complete model of the system (Fig. 10), will allow us to evaluate the relative importance of different sources of uncertainty and also understand how different management decisions (e.g., quota allocation) reverberate through the system. This will allow us to characterize how policy decisions regarding the total catch quota and economic contributions are linked (H3b in Fig. 2).

Hypothesis 5: Flounder mortality from angling and commercial fishing can be predicted from behavior driven by spatial mismatches between abundance and policy along the north-south gradient

Angler behavior: We will develop a sub-model integrating information on angler demographics to predict their fishing activity and reporting behavior for each state, providing a more complete estimate of fishing mortality from the recreational fishery. Currently, the recreational demand model in the existing MSE predicts the magnitude and size composition of the recreational harvest and discards in response to the current regulation at the state level. However, it assumes full compliance with regulations. Using information on knowledge and drivers of engagement with management rules by anglers (gained from Aim 1) we will adjust the recreational demand model accordingly to estimate true mortality.

Commercial fishery: We propose to develop a sub-model integrating economic and policy drivers of commercial fishing activity in each state. Currently, simplifying assumptions about the commercial fishery are made in the MSE. The fishery is assumed to catch its quota regardless of stock size and availability. It ignores the complexity that arises from the allocation of quota among states and the changing distribution of the flounder population. Using the available data on landings, quota allocation, and flounder abundance, as well as data and findings from surveys in Aims 1 and 2, we will construct functional relationships (including their variability) between the economic drivers of demand for flounder, costs of fishing, and actual commercial landings in each state to create a model of commercial fishing pressure that is responsive to social and economic conditions. Our advisory group (See Broader Impacts) will help inform the sets of modeled relationships to include as well as their relative plausibility, for use in MSE model scenarios.

Hypothesis 6: Total population size of flounder can be predicted in an integrated model of climate-driven recruitment and fishing pressure.

Working with the postdoc in year 3, Kindsvater and Fay, will develop the recruitment model within the MSE model of summer flounder population dynamics to include spatial drivers of survival in early life from Aim 3. Previous work shows predicting effectiveness of management shows it depends strongly on future recruitment and distribution of flounder. Incorporating climate as a driver of flounder distribution will affect the fishing behavior models described under H5. The result will be an integrated model of flounder dynamics where the stock size, and age, size, and sex structure are responsive to both spatial complexities in recruitment and in fishing fleet dynamics. The population dynamic modeling in the MSE is similar to models employed by Kindsvater and postdoc Brooks (Brooks et al. *in revision*; Table 2).

Table 2. Equations for age- and size-specific population dynamics in the MSE (modified from Kindsvater et al. 2020).

Process	Equation	Interpretation
Length-at-age function	$L(i, a + 1) = L(i, a)e^{-k} + L_{\infty,i}(1 - e^{-k_i})$	Discrete version of the von Bertalanffy growth equation: $L_{\infty,i}$ and k_i determine size-at-age L (of sex i at age a).
Maturation probability function	$p_m(i, a) = \frac{1}{1 + e^{-q(a - a_{mat,i})}}$	Function describing the cumulative probability (ogive) individuals are mature at age a ; $a_{mat,i}$ is the age at which 50% of individuals mature; q determines steepness of function
Recruitment	Climate- and density-dependent function	Derived from Aim 3
Mortality	$M(i, a) = \mu L(i, a)^b$	Mortality depends on length, which depends on age; μ and b are scaling and shape parameters (Lorenzen 1996).
Population dynamics through time	$N(a + 1, t + 1) = \begin{cases} N_0(t) & \text{if } a = 0 \\ N(a, t)e^{-(M(a) + F(a))} & \text{if } a > 0 \end{cases}$	$N(i, a, t)$ is the number of individuals age a alive at time t . F(a) - fishing mortality - is derived from Aims 1-2

In prior work, uncertainty associated with recruitment was set equal to the level of scientific uncertainty currently assumed by the MAFMC's Scientific and Statistical Committee when applying their risk policy to recommend catch limits; this work will allow us to more mechanistically quantify how much uncertainty we expect in recruitment. We will work with our advisory group identify realistic scenarios for bias and uncertainty associated with key data streams used in summer flounder stock assessments (See Broader Impacts). The advisory group will ensure that we are able to quantify the implications of mis-specifying key processes within the management decision-making process in order to ensure model outputs are useful to decision makers.

Hypothesis 7: Ecosystem services of flounder -economic value, cultural value, and food security increase when disparities between the quota and abundance are minimized

Research design: The integrated simulation model framework will then be applied to a set of scenarios representing combinations of economic drivers, fisher behavior drivers, and fish population dynamics in different climate scenarios to understand how the interaction of these affects the overall performance of different options for managing the summer flounder fishery. In addition to the operating model characteristics, we will determine a set of management alternatives that vary how decisions related to distributing quota (or target catch) to actors (state managers, anglers, and commercial fishers) are made. This will range from using *status quo* approaches based on historical observations of catches and stock distribution, to dynamic allocation rules that update the distribution of quota based on changing availability of flounder to the fishery. These rules may be based on abundance but they could also be based on expected changes to factors of economic or cultural value, identified during work in both Aims 1 and 2.

Running our integrated MSE in different scenarios will allow us to determine if the integrated coastal socio-environmental system is more sustainable when there is better alignment of fishery decision making with the abundance and distribution of the stock. However, we anticipate there may be distinct winners and losers both spatially and between the recreational and commercial components of the fishery under different likely scenarios. Our advisory group will refine the set of sustainability performance metrics currently available to best ensure that the priorities of stakeholder groups are addressed in model outputs.

Broader Impacts

Our broader impacts focus on three key audiences, (1) stakeholders that are engaged in the flounder fishery, especially the decision-making process, and (2) broader members of coastal communities that are affected by changes in the socio-environmental system (discussed below in Year 5 activities). We will engage in knowledge co-production (Norstrom et al. 2020) via the creation of an advisory team of representatives from two entities that currently make decisions about the management of summer flounder, the Mid-Atlantic Fisheries Management Council (MAFMC) and the Atlantic States Fisheries Management Commission (ASFMC), which include recreational and commercial fishers and agency staff. We will also invite representatives from under-represented groups, identified and invited based on the qualitative research in Year 1. Expanding on the collaborative model outlined in Aim 4 (Fay et al. 2023), project personnel, will hold five annual meetings with the advisory team where we will seek their contributions to the research including clearly defined priorities and research design in service to ensuring meaningful outcomes of our research (Fig. 10). Specifically, we will use an iterative, coproduction process that includes transdisciplinary engagement with the mechanisms that drive the dynamics of our system (Wall et al. 2017). Goals include building literacy and shared understanding around strategic decision-making processes and how it can be used within fisheries management, and familiarizing the core group with the models and what types of questions they can help address. Members of the MAFMC and ASFMC have formerly collaborated on use of the summer flounder fishery as a case study for NOAA's Ecosystem Based Management Plan, which was led by co-PI Fay.